The Castle Crown Caper

Whodunnit? · Polynomials & Factoring Name: _____

It was the night before the royal coronation at Castle Quadrant. The guards changed shifts, the torches flickered, and when the trumpets sounded, the **Royal Crown** was missing from its velvet cushion!

Six people were in the castle, each in a different part, and there are six sneaky ways to smuggle a crown out. You are the Royal Detective: find out **who** took it, **where** they were, and **how** they did it.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Lady Wren Royal Advisor	☐ Throne Room	☐ Hidden in a treasure chest
☐ Sir Bram Knight Captain	☐ Drawbridge	☐ Tucked inside a lute
☐ Jester Pip Court Jester	☐ Dungeon	☐ Lowered from a window
☐ Chef Margaux Royal Cook	☐ Royal Kitchen	☐ Smuggled in a laundry basket
☐ Wizard Orrin Court Wizard	☐ Tower Library	☐ Hidden in a suit of armor
☐ Bard Elowen Royal Minstrel	☐ Rose Garden	☐ Carried in a magic hat

The culprit is _____ **in** _____ **using** _____

Show Your Work

The Castle Crown Caper · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Castle Crown Caper

Factor:

$$x^2 + 5x + 6$$

A. $(x - 2)(x - 3)$	Sir Bram was polishing his sword in the Tower Library. The librarian shushed him twice.
B. $(x + 2)(x + 3)$	Lady Wren was rehearsing the coronation speech in the Throne Room before a dozen courtiers.
C. $(x + 5)(x + 6)$	Every laundry basket in the castle was dumped out. Only socks and tablecloths were found.
D. $(x + 1)(x + 6)$	Wizard Orrin was casting a growth spell on the Rose Garden. The petals are still glittering.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Castle Crown Caper

Factor:

$$x^2 - 5x + 6$$

A. $(x - 5)(x - 6)$	The laundry maid says her basket never left the washhouse. Its wicker lid is still tied shut.
B. $(x - 1)(x - 6)$	Jester Pip was juggling scrolls in the Tower Library. The librarian wrote him a scolding note with the time on it.
C. $(x + 2)(x + 3)$	Wizard Orrin was presenting a crystal ball in the Throne Room. The guards watched the whole thing.
D. $(x - 2)(x - 3)$	The treasure chest was opened in front of the guards. It was full of old coins.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Castle Crown Caper

Factor:

$$x^2 + x - 6$$

A. $(x + 2)(x - 3)$	Wizard Orrin was testing the Drawbridge chains with a sparkly spell. The gatekeeper took notes.
B. $(x - 2)(x + 3)$	Sir Bram was guarding the Drawbridge. Every knight on duty saw him.
C. $(x - 1)(x + 6)$	Lady Wren was reading old treaties in the Tower Library. Her name is on the sign-in scroll.
D. $(x + 1)(x - 6)$	The big laundry basket was weighed at sunset. It was light as a feather.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #4

The Castle Crown Caper

Factor:

$$x^2 - x - 6$$

A. $(x + 1)(x - 6)$	Chef Margaux was hunting for a lost recipe in the Tower Library. The dusty shelves show her fingerprints.
B. $(x - 2)(x + 3)$	Wizard Orrin was stirring a bubbling potion in the Royal Kitchen. The kitchen helpers all saw the purple smoke.
C. $(x - 1)(x + 6)$	The guards looked in every laundry basket at the gate. Each held only clean linens.
D. $(x + 2)(x - 3)$	The lute’s hollow body was empty. Only a mouse lived inside.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Castle Crown Caper

Factor:

$$x^2 - 9$$

A. $(x - 3)(x + 3)$	Jester Pip was juggling for the prisoners in the Dungeon. The guard laughed the whole time.
B. $(x - 9)(x + 9)$	The laundry baskets were all stacked empty in the washhouse by noon.
C. $(x - 3)^2$	Wizard Orrin was turning rats into mice in the Dungeon. The jailer is still shaking his head.
D. $(x + 3)^2$	Bard Elowen was tuning a lute in the Tower Library. The echoing notes drew a small crowd.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #6

The Castle Crown Caper

Factor completely:

$$2x^2 + 8x$$

A. $2x(x + 4)$	Rope marks on the window are old and dusty. Nothing was lowered recently.
B. $2(x + 4)$	The castle laundry list matches the number of baskets exactly. None went missing.
C. $2x(x + 8)$	Wizard Orrin was chatting with a talking toad in the Rose Garden. Two gardeners overheard.
D. $2x(x + 2)$	Jester Pip was hiding a rubber chicken among the books in the Tower Library. The librarian caught him red-handed.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Castle Crown Caper

Factor:

$$2x^2 + 7x + 3$$

A. $(2x - 1)(x - 3)$	Sir Bram was studying battle maps in the Tower Library. The map table has his muddy bootprints.
B. $(2x + 1)(x - 3)$	Not one laundry basket was seen leaving the castle gates today. The gate log is clean.
C. $(2x + 1)(x + 3)$	Chef Margaux was frosting the coronation cake in the Royal Kitchen.
D. $(2x + 3)(x + 1)$	Wizard Orrin was practicing a fanfare spell in the Throne Room. The court herald counted every note.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #8

The Castle Crown Caper

Factor:

$$x^2 - 6x + 9$$

A. $(x - 3)^2$	The suits of armor were all checked. Each was empty.
B. $(x - 6)^2$	The washerwomen counted their baskets twice. Each one went home with its owner.
C. $(x + 3)^2$	Wizard Orrin was turning pumpkins into coaches in the Royal Kitchen. The cook's apprentices cheered.
D. $(x - 3)(x + 3)$	Bard Elowen was composing a ballad in the Tower Library. Her inkwell is still wet.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

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Multiply:

$$(x + 2)(x - 5)$$

A. $x^2 + 3x - 10$	Wizard Orrin was feeding the moat monster from the Drawbridge. A passing farmer waved hello.
B. $x^2 - 10$	Lady Wren was signing royal letters in the Tower Library. Her wax seal is still warm.
C. $x^2 - 3x - 10$	Bard Elowen was singing under the roses in the Rose Garden. Three ladies listened.
D. $x^2 - 3x + 10$	The laundry basket in the corridor holds only wet capes. The guard checked twice.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

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Factor completely:

$$3x^2 - 12$$

A. $3(x^2 - 4)$	A tag on the laundry basket shows it was locked in the storeroom all night.
B. $3(x - 2)^2$	Chef Margaux was borrowing a cookbook from the Tower Library. The checkout card has the time.
C. $3(x - 4)(x + 4)$	Wizard Orrin was chatting with a ghost in the Dungeon. The guard heard the whole conversation.
D. $3(x - 2)(x + 2)$	The magic hat is a flat, folded hat. It can’t hide a crown.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.